

PERSHORE, United Kingdom

WMO: 035290

Lat: 52.1483N

Lon: 2.0411W

Elev: 31

StdP: 100.95

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.8	(c) -3.1	(d) -5.8	(e) 2.3	(f) -2.3	(g) -4.5	(h) 2.6	(i) -1.0	(j) 12.3	(k) 8.9	(l) 11.2	(m) 9.3	(n) 1.0	(o) 340	(p) 0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	26.7	18.3	24.6	17.4	22.9	16.7	19.3	24.5	18.3	23.0	17.5	21.5	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	12.5	21.0	16.6	11.8	20.0	15.8	11.2	19.3	55.0	24.5	51.9	23.2	49.3	21.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.9	7.9	DB	-8.4	30.3	2.8	2.3	-10.4	31.9	-12.0	33.3	-13.6	34.5	-15.6	36.2
			WB	-8.1	21.6	3.0	1.9	-10.3	23.0	-12.0	24.1	-13.7	25.1	-15.9	26.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	4.8	5.2	6.7	9.0	12.0	14.9	17.1	16.7	14.3	11.0	7.5	5.0
	DBStd	5.39	3.55	3.33	3.08	2.96	2.94	2.60	2.42	2.36	2.44	3.07	3.55	4.02
	HDD10.0	747	164	137	108	52	13	1	0	0	1	25	88	158
	HDD18.3	2934	421	368	359	280	197	106	54	61	122	228	326	413
	CDD10.0	888	2	2	6	22	74	149	220	209	131	56	13	4
	CDD18.3	34	0	0	0	0	1	4	16	12	2	0	0	0
	CDH23.3	324	0	0	0	1	12	57	161	78	14	2	0	0
	CDH26.7	57	0	0	0	0	0	11	32	13	1	0	0	0
Wind	WSAvg	3.5	4.1	4.1	3.9	3.5	3.3	3.1	3.0	3.1	3.1	3.5	3.5	3.8
Precipitation	PrecAvg	633	58	39	46	47	53	58	46	61	60	50	54	61
	PrecMax	792	105	101	108	144	152	166	112	126	174	126	132	128
	PrecMin	434	10	4	4	4	9	4	5	8	8	4	24	12
	PrecStd	84	25	25	23	28	34	40	29	32	39	29	26	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	14.2	17.0	21.8	24.9	28.2	29.9	28.6	25.3	20.2	15.9	14.1
		MCWB	11.2	11.2	11.2	14.4	16.9	18.7	18.9	19.2	17.5	16.0	13.8	12.1
	2%	DB	12.1	12.3	14.7	18.3	22.3	24.7	27.2	25.3	22.5	18.0	14.7	13.0
		MCWB	10.2	10.1	10.6	12.5	15.9	17.4	18.6	18.1	16.6	14.6	13.0	11.6
	5%	DB	11.0	11.2	13.2	16.2	19.9	22.5	25.0	23.3	20.7	16.8	13.7	12.0
		MCWB	9.4	9.3	9.9	11.4	14.5	16.4	17.6	17.2	15.6	14.0	12.0	10.6
	10%	DB	10.0	10.2	11.9	14.5	17.7	20.6	23.1	21.7	19.2	15.7	12.6	10.9
		MCWB	8.4	8.3	9.2	10.5	13.2	15.5	16.7	16.2	14.9	13.3	11.0	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.7	12.4	14.9	18.1	20.7	21.4	20.1	18.9	16.9	14.4	12.6
		MCDB	12.8	13.4	15.1	20.2	23.3	26.3	27.2	25.9	23.6	19.6	15.4	13.5
	2%	WB	10.6	10.5	11.3	13.2	16.4	18.2	19.3	19.0	17.5	15.5	13.3	11.7
		MCDB	11.8	11.9	13.7	17.1	21.0	22.9	25.0	23.7	20.9	17.3	14.5	12.8
	5%	WB	9.6	9.6	10.4	12.1	15.1	17.0	18.3	18.1	16.6	14.6	12.2	10.8
		MCDB	10.8	11.0	12.6	15.4	18.8	21.2	23.5	21.9	19.4	16.3	13.4	11.9
	10%	WB	8.6	8.6	9.5	10.9	13.9	16.0	17.4	17.2	15.7	13.6	11.2	9.6
		MCDB	9.9	10.0	11.6	14.0	17.0	19.7	21.8	20.5	18.3	15.3	12.4	10.8
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.4	8.1	9.3	9.5	9.6	9.9	9.4	9.1	7.6	6.3	5.7
		MCDBR	6.1	6.9	9.7	12.3	13.2	13.1	13.7	11.8	11.6	7.9	6.0	5.8
		MCWBR	5.5	5.4	6.4	7.4	7.6	6.9	6.4	5.8	6.6	5.4	4.9	5.3
	5% WB	MCDBR	6.0	6.8	8.6	11.0	11.5	11.2	11.8	9.9	9.6	7.0	6.0	5.9
		MCWBR	5.7	5.7	6.1	7.0	7.1	6.6	6.5	5.5	6.1	5.4	5.4	5.6
Clear-Sky Solar Irradiance	taub	0.323	0.339	0.372	0.397	0.402	0.398	0.395	0.382	0.368	0.351	0.326	0.313	
	taud	2.358	2.382	2.287	2.229	2.250	2.292	2.303	2.363	2.402	2.427	2.408	2.334	
	Ebn at Noon	669	774	813	836	849	856	851	843	812	748	664	610	
	Edh at Noon	62	79	105	125	129	125	121	108	93	75	58	53	
All-Sky Solar Radiation	RadAvg	0.76	1.43	2.53	3.89	4.79	5.16	4.87	4.02	3.03	1.73	0.92	0.60	
	RadStd	0.07	0.17	0.31	0.44	0.29	0.54	0.51	0.32	0.25	0.15	0.10	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (97 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page